Design and Development of an AI-Based Collaborative Decision and Action Tracking System

(ES-452: Major Project - Dissertation)

submitted in partial fulfillment of the requirement
for the award of the degree of

Bachelor of Technology
in
Artificial Intelligence & Data Science

Submitted by

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March 2026

BHAGWAN PARSHURAM INSTITUTE OF TECHNOLOGY

VISION OF THE INSTITUTE

- To establish a leading Global Center of Excellence in multidisciplinary education, training and research in the area of Engineering, Technology and Management.
- To produce technologically competent, morally & emotionally strong and ethically sound professionals who excel in their chosen field, practice commitment to their profession and dedicate themselves to the service of mankind.

MISSION OF THE INSTITUTE

- To develop world class Laboratories and other Infrastructure conducive in acquiring latest knowledge and expertise.
- To bridge the knowledge and competency gaps of institute's fresh pass-outs vis-a-vis field requirements.
- To strengthen Industry-Institute Interaction and partnership for imbibing corporate culture amongst our faculty and students.
- To promote research culture among faculty and students enhancing their academic and professional confidence needed to face global challenges.
- To honour commitment towards social and moral values.

BHAGWAN PARSHURAM INSTITUTE OF TECHNOLOGY

PROGRAM OUTCOMES (POs)

PO 1. Engineering Knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2. Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3. Design/development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.

PO 4. Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5. Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6. The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multi-disciplinary settings.

PO 10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multi-disciplinary environments.

PO 12. Life-long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

BHAGWAN PARSHURAM INSTITUTE OF TECHNOLOGY

COURSE OUTCOMES (COs)

CO 1. Apply engineering knowledge and design principles to identify, formulate, and solve a complex problem through a structured project.

CO 2. Conduct systematic investigation and analysis using modern tools and techniques for designing and implementing the project.

CO 3. Demonstrate professional documentation, effective oral communication, and presentation skills through the dissertation.

CO 4. Work ethically as an individual or in a team to manage the project effectively, considering societal, environmental, and sustainability aspects.

BHAGWAN PARSHURAM INSTITUTE OF TECHNOLOGY

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

SDG No.	Goal Title
SDG 1	NO POVERTY
SDG 2	ZERO HUNGER
SDG 3	GOOD HEALTH & WELL-BEING
SDG 4	QUALITY EDUCATION
SDG 5	GENDER EQUALITY
SDG 6	CLEAN WATER & SANITATION
SDG 7	AFFORDABLE & CLEAN ENERGY
SDG 8	DECENT WORK & ECONOMIC GROWTH
SDG 9	INDUSTRY, INNOVATION & INFRASTRUCTURE
SDG 10	REDUCED INEQUALITIES
SDG 11	SUSTAINABLE CITIES & COMMUNITIES
SDG 12	RESPONSIBLE CONSUMPTION & PRODUCTION
SDG 13	CLIMATE ACTION
SDG 14	LIFE BELOW WATER
SDG 15	LIFE ON LAND
SDG 16	PEACE & JUSTICE STRONG INSTITUTIONS
SDG 17	PARTNERSHIPS FOR THE GOALS

BHAGWAN PARSHURAM INSTITUTE OF TECHNOLOGY

Department of Artificial Intelligence & Data Science

Vision

To emerge as a center of excellence, in the field of Computer Science and Engineering & Research, by grooming our pupils with strong conceptual knowledge to enable them as a professional and researcher for the benefit of society.

Mission

1. To deliver comprehensive education in Data Science by integrating core principles of Computer Science with modern analytical and computational tools to build competent and industry-ready professionals.
2. To cultivate self-driven learners capable of recognizing and responding to modern societal and technological needs through research, higher studies, or entrepreneurship.
3. To empower students with critical thinking and problem-solving skills to lead and innovate in Data Science and related fields.
4. To foster meaningful industry engagement for the enrichment of all stakeholders.
5. To instill ethical data practices through the development of professional integrity, social responsibility, and a lifelong commitment to learning in students.

Program Educational Objectives (PEOs)

6. Professional Excellence and Adaptability: Graduates will establish themselves as successful professionals or researchers or entrepreneurs in the field of Data Science by applying analytical, computational, and domain-specific knowledge to solve real-world problems, while continuously learning and adapting to emerging technologies.
7. Ethics, Social Responsibility, and Leadership: Graduates will uphold ethical standards and social responsibility in data handling and decision-making processes while demonstrating leadership and teamwork in multidisciplinary environments.
8. Collaborative Engagement and Social Interaction: Graduates will communicate effectively and collaborate with diverse teams, nurturing productive engagement with the community.

Program Specific Outcomes (PSOs)

9. Apply core concepts of Computer Science and Data Science to develop effective and scalable solutions by leveraging data collection, processing, visualization, statistical modelling, and machine learning for real-world problems.
10. Design and develop sustainable data-driven intelligent systems using engineering principles and modern tools, while embracing lifelong learning to adapt to emerging technologies.
11. Graduates will demonstrate ethical and legal awareness in data usage, along with strong communication skills, enabling them to succeed as professionals, entrepreneurs, or in higher studies.

CO-PO MAPPING TABLE

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	1	1	1	1	1	1	1	2
CO2	2	3	3	2	3	1	1	0	1	1	2	2
CO3	1	2	1	1	2	0	0	1	2	3	1	2
CO4	0	0	0	0	0	3	3	3	2	1	0	1

JUSTIFICATION FOR CO-PO MAPPING

CO1 - Apply knowledge and design principles

- PO1-PO3: Core application of knowledge, problem analysis, and design of solutions.
- PO4: Involves investigation during design/implementation.
- PO5: Moderate use of modern tools in solving the problem.
- PO6-PO8: Students are expected to be aware of professional and ethical aspects.
- PO9: Projects typically involve teamwork.
- PO10-PO12: Requires documentation, time management, and learning adaptability.

CO2 - Investigation and implementation

- PO1-PO3: Applies existing engineering principles for practical implementation.
- PO4: High involvement of experimental analysis or modeling.
- PO5: Strong use of modern tools like simulation software, CAD, programming environments, etc.
- PO6-PO7: Considerations may arise in the project scope.
- PO9-PO12: Collaborative work, communication, and continual improvement are essential.

CO3 - Communication and documentation

- PO1-PO2: Applies existing engineering principles for documentation.
- PO3-PO4: Basic alignment as solutions are documented.
- PO5: Tools may be used in technical writing and presentations.
- PO8: Ethical and professional writing required.

- PO9-PO10: High relevance for teamwork and communication.
- PO11-PO12: Documenting budgets, schedules, and reflecting learning outcomes.

CO4 - Ethics, teamwork, sustainability

- PO6: Projects often touch on societal impact or industrial relevance.
- PO7: High relevance if sustainability is part of project consideration.
- PO8: Strong emphasis on ethics and professional conduct.
- PO9: Effective teamwork is a key requirement.
- PO12: Requires self-motivated, lifelong learning to complete capstone work.

CO-PSO MAPPING TABLE

	PSO1	PSO2	PSO3
CO1	3	2	2
CO2	2	3	3
CO3	1	2	3
CO4	1	2	3

JUSTIFICATION FOR CO-PSO MAPPING

CO1 - Apply engineering knowledge and design principles

- PSO1 (3): Strong focus on software/system design using structured principles and algorithms.
- PSO2 (2): Project often requires selecting a working environment (tools, OS, platforms).
- PSO3 (2): Many major projects involve real-world applications using emerging technologies.

CO2 - Investigate and implement project using tools

- PSO1 (2): Implements optimized designs and algorithms.
- PSO2 (3): Direct use and configuration of hardware/software/networking resources.
- PSO3 (3): Strong overlap with modern tech like AI, Cloud, IoT, etc., used in implementations.

CO3 - Document and present findings

- PSO1-PSO3 (1 each): While CO3 is focused on communication, the content presented will be technical, hence a minor relation to each PSO.

CO4 - Ethics, teamwork, sustainability

- PSO2 (2), PSO3 (2): Includes selecting responsible tech environments, use of eco-efficient solutions.
- PSO1 (1): Ethical software development and collaboration influence design decisions.

PROJECT TO SDG MAPPING

PROJECT TITLE: Design and Development of an AI-Based Collaborative Decision and Action Tracking System

MAPPED SUSTAINABLE DEVELOPMENT GOALS: 4 (Quality Education), 8 (Decent Work & Economic Growth), 9 (Industry, Innovation & Infrastructure), 12 (Responsible Consumption & Production), 16 (Peace, Justice & Strong Institutions), 17 (Partnerships for the Goals)

Justification for SDG mapping:

SDG 4: Quality Education
The proposed system enhances collaborative learning by helping student teams convert discussions into structured outcomes such as summaries, decisions, and action items. It supports academic environments by improving clarity in group work, project planning, and knowledge sharing. By integrating AI-based analysis into collaboration, the system contributes to modern educational practices and promotes effective learning and teamwork.

SDG 8: Decent Work & Economic Growth
Efficient collaboration is essential in professional environments. This project aims to improve team productivity by reducing time spent on unstructured discussions and enabling clear decision-making. By providing actionable insights from conversations, the system helps individuals and teams work more efficiently, thereby supporting productivity and better workplace practices.

SDG 9: Industry, Innovation & Infrastructure
The project contributes to innovation by developing an intelligent system that integrates AI and NLP techniques into collaborative platforms. It introduces a structured approach to analyzing discussions and extracting meaningful insights, thereby supporting the development of smarter digital collaboration tools and advancing technological infrastructure in communication systems.

SDG 12: Responsible Consumption & Production

The system promotes responsible use of information by organizing and summarizing discussions to avoid redundancy and miscommunication. By providing clear insights and structured outputs, it reduces inefficiencies and ensures better utilization of time and digital resources in collaborative environments.

SDG 16: Peace, Justice & Strong Institutions

Clear communication and transparency are critical for decision-making in organizations. This project supports structured and accountable decision-making by tracking discussions and highlighting unresolved issues. This improves transparency, reduces ambiguity, and supports fair and well-informed decisions in collaborative settings.

SDG 17: Partnerships for the Goals

The system encourages effective collaboration among individuals and teams by improving communication and coordination. It can be used across academic groups, project teams, and organizations, thereby promoting partnerships and teamwork through better alignment and shared understanding.

DECLARATION

This is to certify that the material embodied in this Major Project - Dissertation titled

"Design and Development of an AI-Based Collaborative Decision and Action Tracking System"

submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence & Data Science is based on my original work. It is further certified that this Major Project - Dissertation work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma. My indebtedness to other works has been duly acknowledged at the relevant places.

(Alokit Mishra)
01720811922

CERTIFICATE

This is to certify that the work embodied in this Major Project - Dissertation titled

"Design and Development of an AI-Based Collaborative Decision and Action Tracking System"

submitted in the partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Artificial Intelligence & Data Science, is original and has been carried out by Alokit Mishra (Enrollment No. 01720811922) under my supervision and guidance.

It is further certified that this Major Project - Dissertation work has not been submitted in full or in part to this university or any other university for the award of any other degree or diploma to the best of my knowledge and belief.

(Dr. Suman Arora)
Assistant Professor, Department of CSE

ACKNOWLEDGEMENT

An endeavor is not complete and successful till the people who made it possible are given due credit for making it possible. I take this opportunity to thank all those who have made the endeavor successful for me.

At the very onset I thank Dr. Suman Arora, my Major Project - Dissertation supervisor for giving inspiration on such important and valuable topic, her scholarly guidance, constant supervision and encouragement to make it a success. Her research knowledge and passion for problem solving amazes and inspires me. At all crucial stages, valuable insights given by her made me take the right direction. I thank her for the countless hours she has spent with me, criticizing my ideas, enlightening my writing skills, and helping me. Her assistance during this work has been invaluable and inspirational.

I extend my thanks to Prof. Achal Kaushik, HOD-CSE, BPIT and Prof. Monika Arora, In-charge AIDS & CSE-DS for their regular motivation, support and prayerful presence.

Last but not least, I especially thank my family members for their unconditional moral support, encouragement, best wishes and patience for not giving them proper time during the study.

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ABSTRACT

This project presents an AI-Based Collaborative Decision and Action Tracking System, designed to analyze group discussions and automatically generate structured outputs such as summaries, key decisions, action items, and unresolved issues. The system aims to enhance team productivity by converting unstructured communication into actionable insights using Natural Language Processing (NLP) techniques.

In modern academic and professional environments, collaboration is primarily conducted through digital communication platforms such as group chats and discussion forums. However, these platforms often fail to provide structured outcomes from discussions, leading to loss of important decisions, unclear responsibilities, and inefficient task management. Extracting meaningful insights from unstructured conversations remains a significant challenge.

The proposed system consists of a web-based interface that allows users to create teams, participate in discussions, and analyze conversations. The backend processes textual data and applies NLP techniques for summarization, keyword extraction, and classification of sentences into categories such as decisions, actions, and general discussion. The system also maintains a structured database of discussions and results, enabling users to track progress and responsibilities.

To evaluate the effectiveness of the system, multiple performance indicators such as accuracy of extracted decisions, relevance of summaries, and user-level interaction metrics are considered. The system is designed to work efficiently for small-scale team collaboration scenarios such as academic project groups and organizational teams.

The implementation demonstrates how AI techniques can be applied to real-world collaboration problems, providing a practical solution for improving communication clarity and decision-making efficiency. The results highlight the potential of integrating AI-driven analysis into collaborative platforms to enhance productivity, reduce ambiguity, and support structured team workflows.

TABLE OF CONTENTS

Page No. (in Roman)

Vision and Mission of the Institute	I
Program Outcomes (POs)	II
Course Outcomes (COs)	IV
Sustainable Development Goals (SDGs)	V
Department's Vision, Mission, PEOs and PSOs	VI
CO-PO and CO-PSO Mapping	VII
Project to SDG Mapping	X
Declaration	XII
Certificate	XIII
Acknowledgement	XIV
Abstract	XV
Table of Contents	XVI
List of Figures	XVII
List of Tables	XVII

Page No. (in Numeric)

Chapter 1: Introduction	1
1.1 Overview	1
1.2 Background and Motivation	2
Chapter 2: Problem Statement	6
2.1 Problem Definition	6
2.2 Objectives	7
Chapter 3: Analysis and Literature Review	9
3.1 Collaborative Systems and Communication Platforms	9
3.2 Natural Language Processing in Text Analysis	10
3.3 Information Extraction and Decision Support Systems	10
3.4 Challenges in Collaborative Data Analysis	11
3.5 Technologies Used	11
3.6 Relevance to Proposed System	12
Chapter 4: Design and Methodology	13
4.1 Methodology Used	13
4.2 Data Handling and Processing	16
4.3 Tools and Technologies Used	17
4.4 System Workflow	18
4.5 Evaluation Approach	18
Chapter 5: Implementation and System Development	19
5.1 Data Presentation	19
5.2 Implementation of the Work	21
5.3 Analysis and Interpretation	23

Chapter 6: Results and Discussion 24
6.1 Results 24
6.2 Discussion 25
6.3 Findings 27
Chapter 7: Summary and Conclusion 28
Chapter 8: Limitation of the Project and Future Work 29
8.1 Limitations of the Project 29
8.2 Future Work 30
Bibliography 32

LIST OF FIGURES

Figure No.	Figure Title	Page No.
Fig 4.1	System Architecture Diagram	12
Fig 4.2	Use Case Diagram	12
Fig 4.3	Activity Diagram of Discussion Flow	13
Fig 4.4	ER Diagram of the System	14
Fig 5.1	Login Page Interface	17
Fig 5.2	Team Discussion Interface	19
Fig 5.3	Analysis Output (Summary & Decisions)	20

LIST OF TABLES

Table No.	Table Title	Page No.
Table 3.1	Functional Requirements of the System	9
Table 3.2	Non-Functional Requirements of the System	10
Table 6.1	Test Cases and Results	24

CHAPTER 1: Introduction

1.1 Overview

In recent years, digital collaboration has become an essential part of academic and professional environments. Teams frequently communicate through online platforms such as messaging applications and discussion forums to share ideas, coordinate tasks, and make decisions. However, these discussions are often unstructured, leading to loss of important information such as decisions, responsibilities, and pending tasks. This creates confusion, reduces productivity, and makes it difficult to track progress within teams.

Existing collaboration platforms primarily focus on enabling communication rather than ensuring decision clarity. While these tools allow users to exchange messages efficiently, they do not provide mechanisms to automatically extract meaningful insights such as summaries, key decisions, or action items from conversations. As a result, users are required to manually interpret discussions, which is time-consuming and prone to errors.

With the advancement of Artificial Intelligence and Natural Language Processing (NLP), it is now possible to analyze textual data and extract structured information from unstructured content. NLP techniques can be used to identify patterns, classify sentences, and generate summaries from large volumes of text. These capabilities provide an opportunity to enhance collaboration systems by integrating intelligent analysis features that assist users in decision-making and task management.

This project introduces an AI-Based Collaborative Decision and Action Tracking System, which aims to bridge the gap between communication and decision-making. The system is designed to analyze group discussions and automatically extract structured outputs such as summaries, key decisions, assigned action items, and unresolved issues. By converting unstructured conversations into organized information, the system improves clarity, accountability, and efficiency in team workflows.

The project is structured to demonstrate both system-level design and intelligent data processing. It involves multiple components including requirement analysis, system

architecture design, implementation of discussion modules, integration of NLP-based analysis, and evaluation of system performance. The development approach ensures that the system is modular, scalable for small teams, and capable of delivering meaningful outputs from real-world conversational data.

1.2 Background and Motivation

The rapid growth of digital communication has transformed the way individuals and teams collaborate. In both academic and professional settings, most interactions now take place through online platforms such as messaging applications and discussion tools. While these platforms enable fast and convenient communication, they often lack mechanisms to organize and interpret the information exchanged during discussions.

A major challenge in collaborative environments is the inability to track decisions and responsibilities effectively. Important conclusions are often buried within long conversation threads, making it difficult for team members to recall key outcomes. This leads to repeated discussions, missed tasks, and reduced overall efficiency. The absence of structured insights from discussions creates a gap between communication and actionable results.

1.2.1 Background

The background of this project lies in the increasing reliance on artificial intelligence systems for processing and interpreting textual information. With advancements in artificial intelligence and natural language processing, it has become possible to analyze textual data and extract meaningful information automatically. These technologies provide an opportunity to enhance existing collaboration systems by introducing intelligent features that assist users in understanding and managing discussions more effectively.

1.2.2 Motivation

The motivation behind this project is to address the challenges of unstructured digital communication by developing a system that bridges the gap between communication and

decision-making. The proposed system aims to simplify the process of extracting useful information from discussions and presenting it in a structured manner. By doing so, it helps users focus on outcomes rather than navigating through large volumes of unorganized text. The system has practical relevance in both academic project teams and professional organizational settings.

CHAPTER 2: Problem Statement

2.1 Problem Definition

In modern academic and professional environments, teams frequently collaborate using digital communication platforms such as group chats, discussion forums, and messaging applications. These platforms enable fast and convenient communication; however, they often fail to produce structured outcomes from discussions. Important information such as key decisions, assigned responsibilities, and pending tasks is often lost within long conversation threads.

Despite the availability of advanced collaboration tools, most existing systems primarily focus on enabling communication rather than extracting meaningful insights from discussions. Users are required to manually interpret conversations to identify decisions and action items, which is time-consuming and prone to errors. This lack of structured information leads to inefficiencies, miscommunication, and reduced productivity within teams.

With the increasing use of Artificial Intelligence and Natural Language Processing (NLP), it is now possible to analyze textual data and extract useful information from unstructured content. However, most existing solutions do not effectively integrate these capabilities into collaborative systems in a way that is both practical and user-friendly.

The core challenges addressed by this project include:

12. Lack of Structured Outcomes from Discussions: Conversations in collaborative environments are unstructured, making it difficult to identify key decisions and action items.
13. Manual Effort in Information Extraction: Users must manually read and interpret discussions to extract useful information, which is inefficient and error-prone.
14. Limited Integration of AI in Collaboration Tools: Existing platforms do not effectively utilize AI techniques to enhance decision-making and information extraction.

15. Difficulty in Managing Large Conversations: As the volume of messages increases, it becomes challenging to summarize discussions and extract meaningful insights.

In light of these challenges, the core problem addressed by this project can be summarized as follows: There is a need for an intelligent system that can automatically analyze group discussions and convert unstructured conversations into structured outputs such as summaries, decisions, and action items to improve collaboration and productivity.

2.2 Objectives

The primary objective of this project is to design and develop an intelligent system that enhances collaboration by converting unstructured discussions into meaningful and actionable outputs. The system aims to assist users in identifying key decisions, tracking responsibilities, and improving overall productivity.

2.2.1 Primary Objectives

16. Implement AI-Based Text Analysis: To apply Natural Language Processing techniques to analyze discussion content and extract summaries, key decisions, and action items.
17. Identify and Track Actionable Insights: To automatically detect important information such as decisions, assigned tasks, and unresolved issues within conversations.
18. Improve Team Productivity and Decision-Making: To reduce ambiguity in discussions and enable users to focus on actionable outcomes rather than raw conversation data.
19. Develop a Collaborative Discussion Platform: To design a web-based system that allows users to create teams, participate in discussions, and manage conversation data effectively.

2.2.2 Secondary Objectives

- Design a Modular and Scalable System Architecture: To ensure that the system can be extended with additional features such as advanced NLP models and integration with external tools.
- Enhance User Experience: To develop an intuitive interface that allows users to easily interact with the system and access insights.
- Evaluate System Effectiveness: To assess the performance of the system based on accuracy of extracted information, relevance of summaries, and usability.
- Support Real-World Collaboration Scenarios: To ensure that the system can be used effectively in academic project teams and small organizational environments.
- Enable Future Enhancements: To provide a foundation for future improvements such as real-time analysis, advanced AI models, and integration with existing communication platforms.

CHAPTER 3: Analysis and Literature Review

The rapid growth of digital collaboration platforms has significantly changed the way individuals and teams communicate and work together. Modern tools such as messaging applications, discussion forums, and project management systems enable seamless communication, but they often lack the ability to extract structured information from conversations. This limitation has led researchers to explore the integration of Artificial Intelligence and Natural Language Processing (NLP) techniques into collaborative systems.

This chapter reviews the existing research and technologies related to collaborative platforms, natural language processing techniques, and intelligent information extraction systems. It also highlights how these concepts have influenced the design and development of the proposed system.

3.1 Collaborative Systems and Communication Platforms

Collaboration platforms such as Slack, Microsoft Teams, and Discord are widely used for communication in both academic and professional environments. These systems allow users to exchange messages, share files, and coordinate tasks efficiently. However, they primarily focus on communication rather than extracting meaningful insights from discussions.

Research studies indicate that users often struggle to track decisions and action items in long conversation threads. Important information is frequently lost or overlooked, leading to inefficiencies and repeated discussions. This highlights the need for intelligent systems that can automatically organize and interpret communication data.

3.2 Natural Language Processing in Text Analysis

Natural Language Processing (NLP) plays a crucial role in analyzing and understanding textual data. NLP techniques such as tokenization, sentence classification, keyword

extraction, and text summarization are widely used to process unstructured text and convert it into structured information.

Recent advancements in NLP have enabled systems to automatically generate summaries, identify key phrases, and classify sentences based on their meaning. These capabilities are particularly useful in collaborative environments where large volumes of text are generated during discussions. The proposed system utilizes NLP techniques to extract important information such as summaries, decisions, and action items from conversation data.

3.3 Information Extraction and Decision Support Systems

Information extraction is a process of identifying structured data from unstructured text. In collaborative systems, this involves detecting important elements such as tasks, decisions, and responsibilities within discussions.

Decision support systems use extracted information to assist users in making informed decisions. By combining information extraction with visualization techniques, such systems can provide meaningful insights and improve productivity. The proposed project integrates these concepts by developing a system that automatically extracts and presents actionable insights from discussions.

3.4 Challenges in Collaborative Data Analysis

Despite advancements in AI and NLP, several challenges remain in analyzing collaborative discussions:

- Unstructured Nature of Conversations: Discussions are often informal and lack a fixed structure, making it difficult to extract meaningful information.
- Implicit Information: Decisions and responsibilities are not always explicitly stated, requiring contextual understanding.
- Large Volume of Data: As discussions grow, processing and summarizing information becomes more complex.

- Accuracy of Extraction: Ensuring that extracted summaries and decisions are accurate and relevant is a major challenge.

These challenges highlight the need for intelligent systems that can effectively process and interpret conversational data.

3.5 Technologies Used

Table 3.1: Functional Requirements of the System

Requirement	Description
User Registration & Login	Users can register and log into the system securely.
Team Creation	Users can create and manage discussion groups/teams.
Discussion Interface	Users can post and view messages within a team.
Trigger Analysis	Users can initiate NLP analysis on a discussion.
Summary Generation	System generates concise summaries of discussions.
Decision Extraction	System identifies and displays key decisions.
Action Item Detection	System extracts and lists assigned tasks.
Result Dashboard	Structured output is displayed on a dashboard.

Table 3.2: Non-Functional Requirements of the System

Requirement	Description
Performance	System should respond to analysis requests within 5 seconds.
Usability	Interface must be intuitive and easy to navigate.
Reliability	System should handle errors gracefully without data loss.
Scalability	Architecture should support future expansion of users and features.
Security	User data and discussions must be stored securely.
Maintainability	Code should be modular and well-documented for easy updates.

The system is built using the following technology stack: Frontend using HTML, CSS, JavaScript/React for user interface; Backend using Python with FastAPI framework; Database using SQLite/PostgreSQL for storing user data and discussions; NLP Libraries

including spaCy, NLTK, and transformer-based models; Development Tools including VS Code, Git and GitHub for version control.

3.6 Relevance to Proposed System

The reviewed literature and technologies provide a strong foundation for the proposed system. By combining collaborative platforms with NLP-based analysis, the project aims to address the limitations of existing systems. The system focuses on transforming unstructured discussions into structured outputs, thereby improving decision-making, task tracking, and overall productivity. The integration of AI techniques ensures that the system is capable of providing meaningful insights from real-world conversational data.

CHAPTER 4: Design and Methodology

4.1 Methodology Used

The proposed project follows a structured development methodology designed to build an intelligent system capable of analyzing collaborative discussions and extracting meaningful insights. The methodology is based on a modular system design approach, where different components such as user interaction, data processing, and analysis are developed and integrated step by step.

The system is designed as a web-based application that processes textual discussion data and applies Natural Language Processing (NLP) techniques to generate structured outputs. The overall methodology consists of multiple stages including system design, data handling, text analysis, and result visualization.

The development process can be divided into the following stages:

4.1.1 System Design and Architecture

In the first stage, the overall architecture of the system is designed. The system follows a client-server architecture consisting of three main components:

- Frontend Layer: Responsible for user interaction and displaying results
- Backend Layer: Handles data processing and application logic
- Analysis Layer: Performs NLP-based processing and extraction

The architecture ensures modularity, scalability, and ease of integration of additional features.

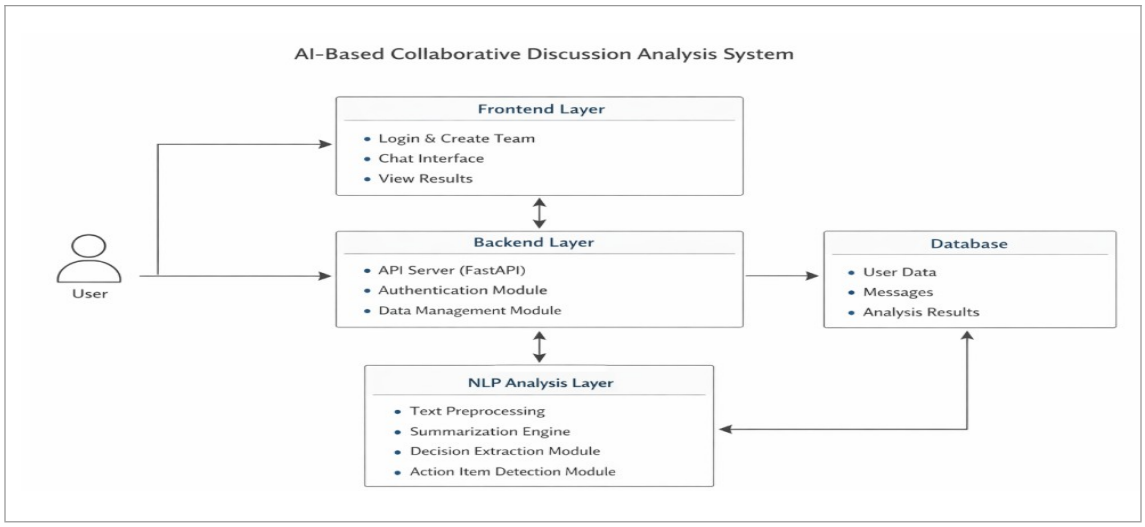

Fig 4.1: System Architecture Diagram

4.1.2 User Interaction Module

This module allows users to interact with the system through a web interface. The key functionalities include:

- User registration and login
- Creation of teams and discussion groups
- Posting and viewing messages
- Initiating analysis of discussions

The interface is designed to be simple and intuitive, ensuring ease of use for users.

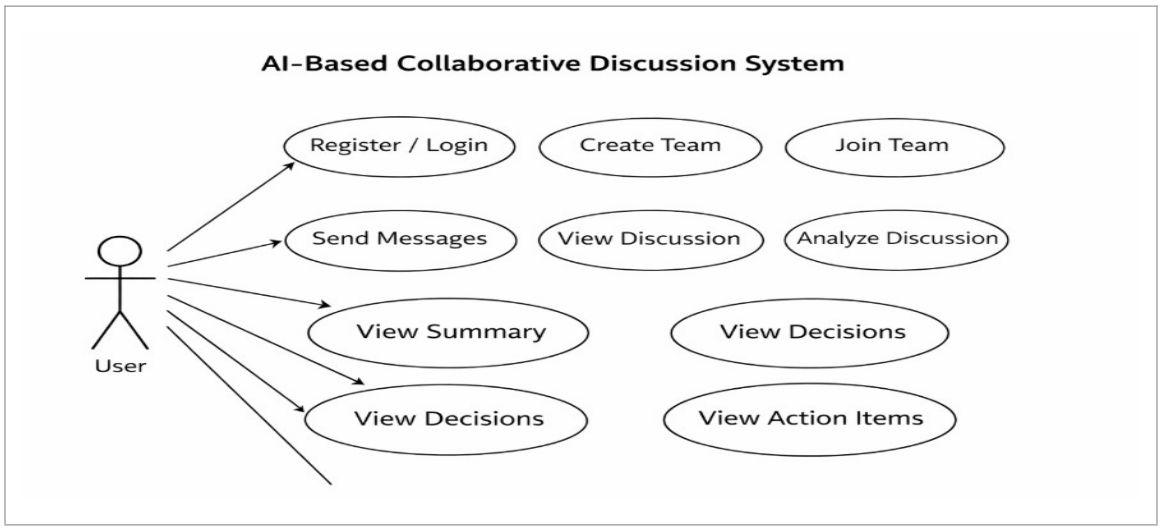

Fig 4.2: Use Case Diagram

4.1.3 Data Processing Module

In this stage, the system processes raw textual data from user discussions. The processing steps include:

- Text cleaning and preprocessing
- Tokenization of sentences
- Removal of irrelevant data
- Structuring messages for analysis

This step prepares the data for further NLP-based processing.

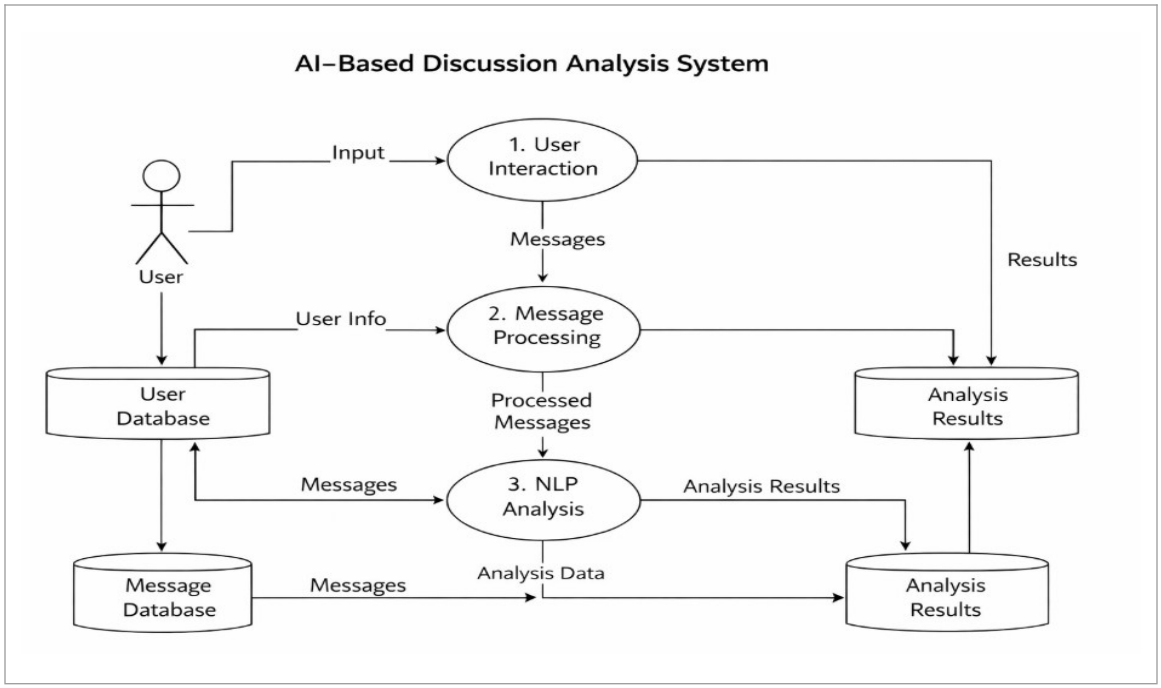

Fig 4.3: Activity Diagram of Discussion Flow

4.1.4 NLP-Based Analysis Module

The core functionality of the system is implemented in this module. It applies Natural Language Processing techniques to extract meaningful insights from discussions. The analysis includes:

- Text Summarization: Generating concise summaries of discussions
- Decision Extraction: Identifying key decisions from conversations

- Action Item Detection: Extracting tasks and responsibilities
- Keyword Identification: Highlighting important topics

The system uses a combination of rule-based and model-based approaches to improve accuracy.

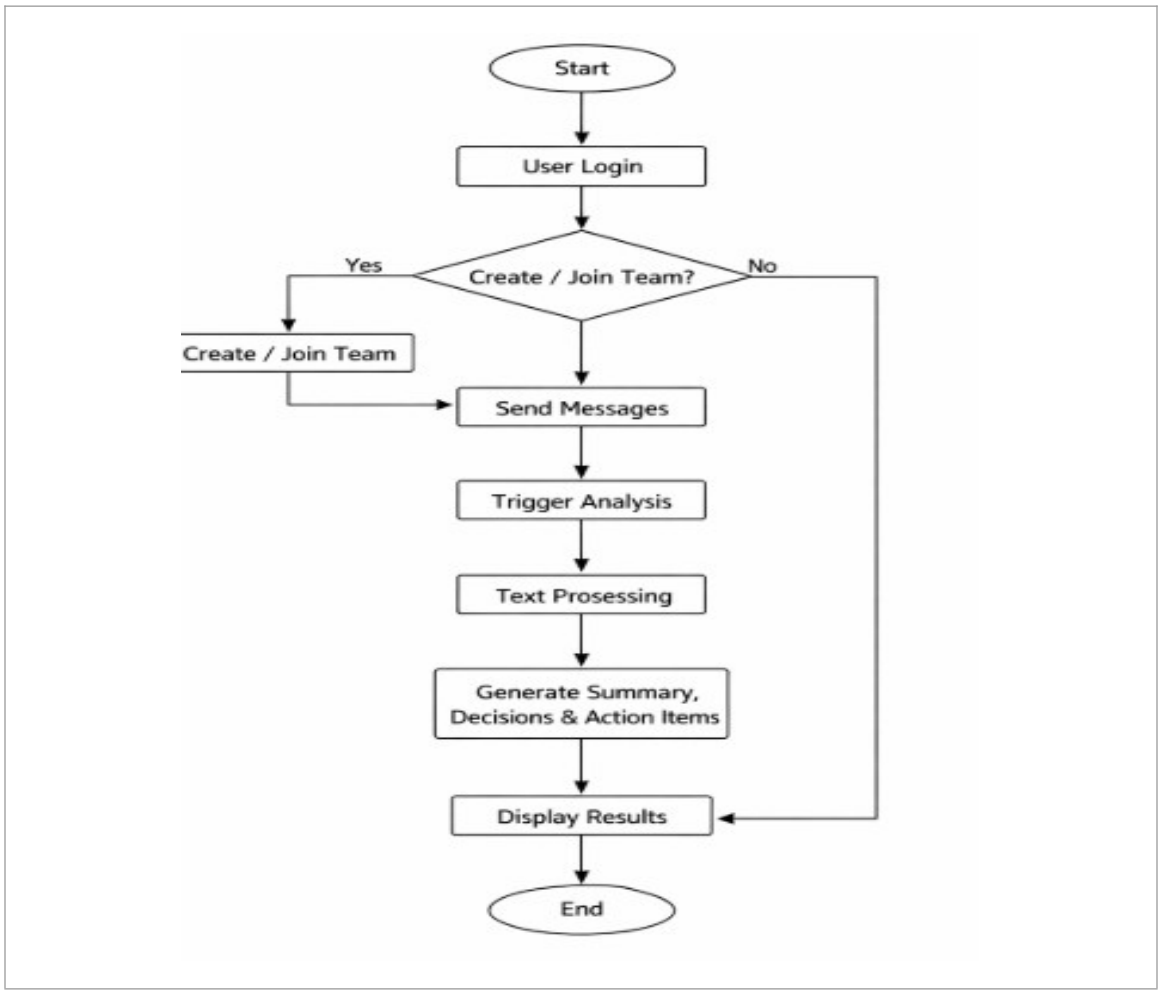

Fig 4.4: ER Diagram of the System

4.1.5 Result Visualization Module

The extracted information is presented to users in a structured format through a dashboard. This includes:

- Summary of discussions
- List of decisions
- Action items with assigned users

- Unresolved or pending issues

Visualization improves usability and helps users quickly understand discussion outcomes.

4.2 Data Handling and Processing

The system uses discussion data generated by users as the primary input. This data is processed and stored in a structured format in the database. The data handling process includes:

- Storing user and team information
- Managing discussion messages
- Storing analysis results
- Retrieving data for visualization

The system ensures efficient data management and retrieval for smooth performance.

4.3 Tools and Technologies Used

4.3.1 Programming Language

- Python (Backend development and NLP processing)
- JavaScript (Frontend development)

4.3.2 Frameworks and Libraries

- FastAPI (Backend API development)
- React / HTML-CSS-JS (Frontend interface)
- spaCy / NLTK / Transformers (NLP processing)

4.3.3 Database

- SQLite / PostgreSQL for storing user data and discussions

4.3.4 Development Tools

- Visual Studio Code

- Git and GitHub for version control

4.4 System Workflow

The working of the system can be described in the following steps:

20. User logs into the system
21. User creates or joins a discussion group
22. Users exchange messages within the group
23. User initiates analysis of the discussion
24. System processes the text using NLP techniques
25. Extracted results are displayed on the dashboard

This workflow ensures a smooth and efficient process from data input to result generation.

4.5 Evaluation Approach

The system is evaluated based on the following parameters:

- Accuracy of extracted decisions and action items
- Relevance and clarity of generated summaries
- User interface usability
- Response time of the system

These factors help in assessing the effectiveness of the system in real-world scenarios.

CHAPTER 5: Implementation and System Development

5.1 Data Presentation

The proposed system uses textual discussion data as the primary input for analysis. This data is generated through user interactions within the collaborative platform. Each discussion consists of multiple messages exchanged between users in a group environment.

The system processes the following components:

- User Messages: Text entered by users during discussions
- User Information: Details of users participating in the discussion
- Discussion Context: Group-based conversations and message history

Each message is treated as an individual input unit and is processed to extract meaningful information. The system converts raw discussion text into structured outputs using Natural Language Processing techniques.

For demonstration and testing purposes, sample discussion datasets were created to simulate real-world conversations. These datasets include multiple participants, varied message lengths, and different types of communication patterns such as decision-making, task assignment, and general discussion.

The dataset is divided into:

- Input Data: Raw discussion messages entered by users
- Processed Data: Cleaned and structured text used for analysis
- Output Data: Extracted summaries, decisions, and action items

This structure ensures that the system can handle real-world conversational data effectively.

5.2 Implementation of the Work

The implementation of the project follows a modular approach, where different components of the system are developed and integrated step by step. The system is implemented as a web-based application consisting of frontend, backend, and analysis modules.

5.2.1 User Interface Implementation

The frontend of the system provides an interactive interface for users. The following pages have been developed:

- Login and registration page
- Team creation and management interface
- Discussion interface for message exchange
- Analysis output page displaying results

The user interface is designed to be simple and intuitive, allowing users to easily interact with the system and view results.

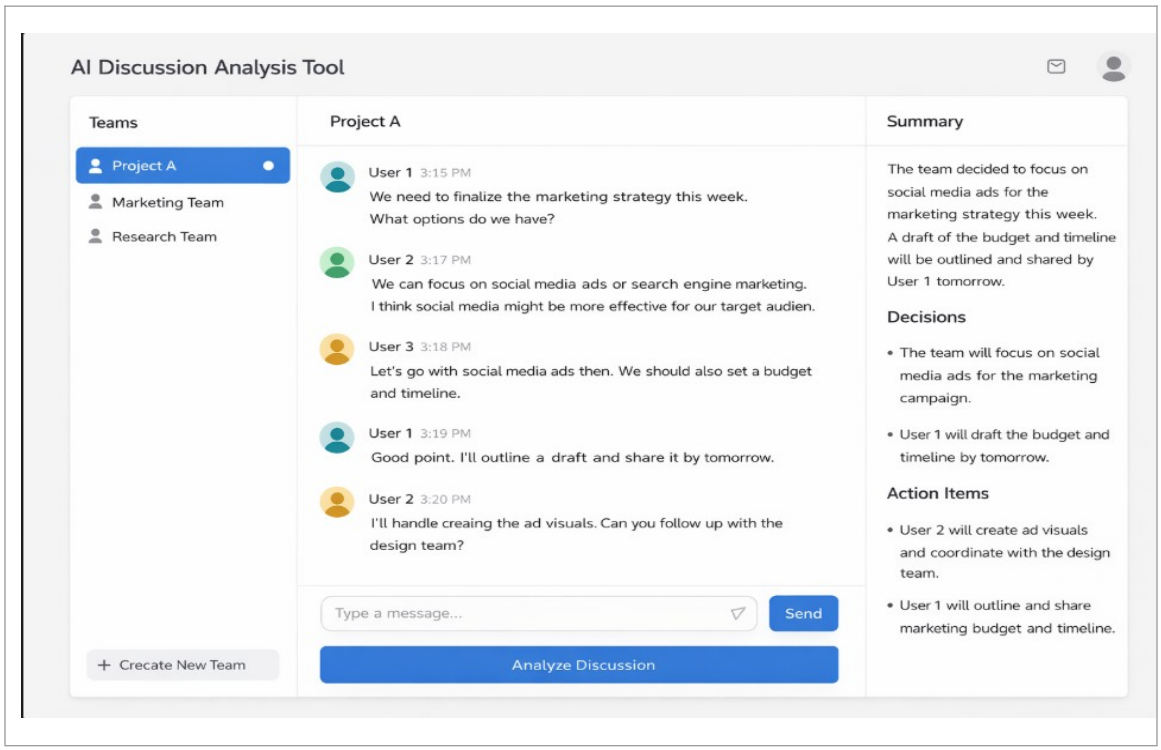

Fig 5.1: Login Page Interface

5.2.2 Backend Implementation

The backend is responsible for handling user requests, processing data, and managing the application logic. It includes:

- API endpoints for user actions
- Data storage and retrieval
- Communication between frontend and analysis module

The backend is implemented using Python-based frameworks to ensure flexibility and efficiency.

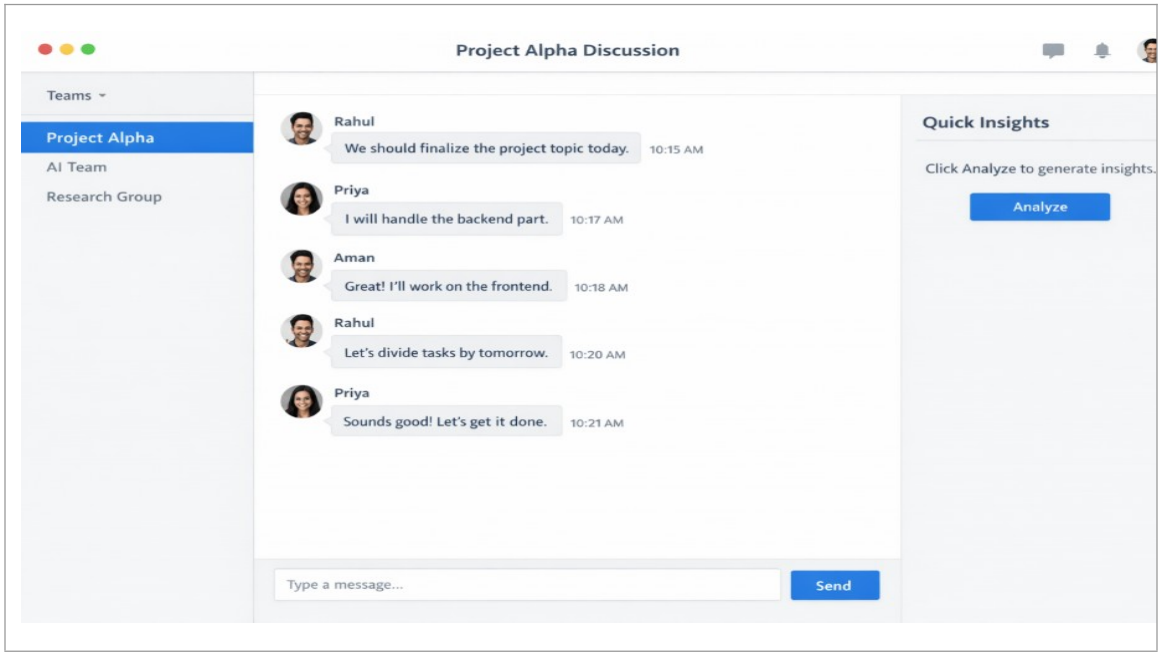

Fig 5.2: Team Discussion Interface

5.2.3 NLP-Based Analysis Implementation

The core functionality of the system is implemented using Natural Language Processing techniques. The system performs the following tasks:

- Text preprocessing (cleaning and tokenization)
- Sentence classification (decision, action, discussion)
- Text summarization
- Keyword extraction

The analysis module processes the input discussion and generates structured outputs such as summaries, decisions, and action items.

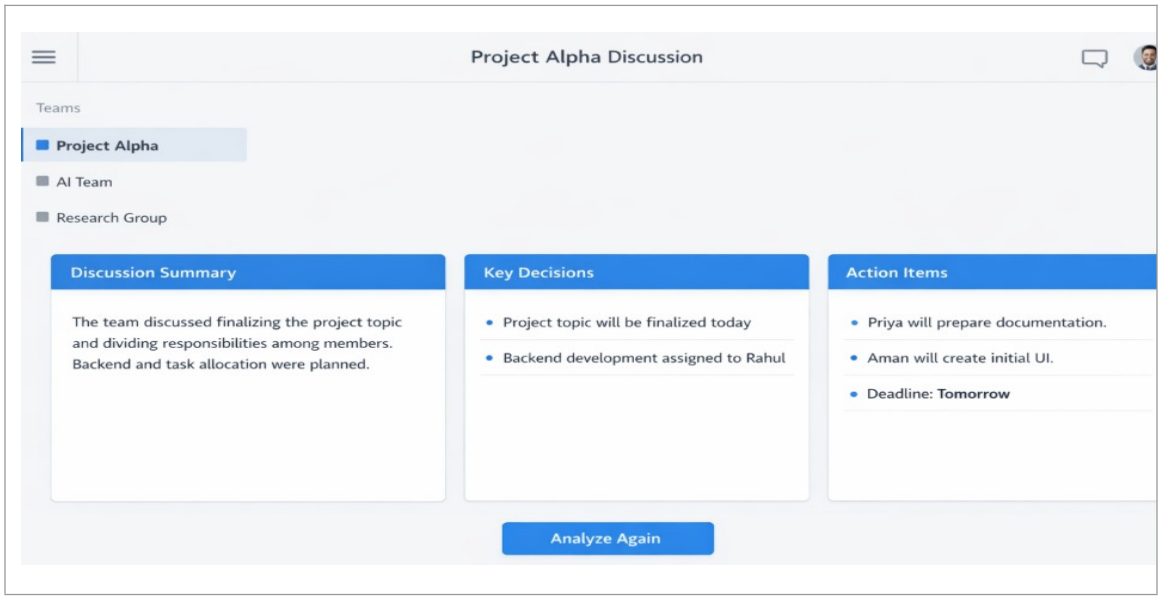

Fig 5.3: Analysis Output (Summary & Decisions)

5.2.4 Integration of System Modules

All modules are integrated to create a complete workflow:

26. User inputs discussion messages
27. Data is sent to the backend
28. NLP module processes the text
29. Results are generated
30. Output is displayed on the user interface

This integration ensures smooth communication between system components.

5.2.5 Prototype Development

A working prototype of the system has been developed to demonstrate the core functionality. The prototype includes:

- Basic discussion interface
- Analysis trigger functionality
- Display of extracted results

This prototype provides a foundation for further development and enhancement.

5.3 Analysis and Interpretation

The system was tested using sample discussion data to evaluate its performance and effectiveness. The results demonstrate that the system is capable of extracting meaningful insights from unstructured conversations.

5.3.1 Discussion Analysis Results

The system successfully generates summaries of discussions, reducing the need for users to manually read long conversation threads. The summaries capture the key points and overall context of the discussion.

5.3.2 Decision and Action Extraction

The system identifies important decisions and action items within conversations. These outputs help users understand responsibilities and tasks more clearly.

5.3.3 Observations

The following observations were made during testing:

- The system effectively reduces information overload in discussions
- Extracted summaries improve readability and understanding
- Decision tracking helps in improving accountability
- The accuracy of extraction depends on the clarity of input messages

5.3.4 Key Findings

The implementation demonstrates that integrating AI techniques into collaborative platforms can significantly improve productivity and communication clarity. Even with a basic prototype, the system shows potential for real-world applications in academic and professional environments.

CHAPTER 6: Results and Discussion

6.1 Results

After completing the development and implementation of the system, the performance was evaluated using sample discussion datasets. The evaluation focused on the system's ability to extract meaningful insights such as summaries, decisions, and action items from unstructured conversations.

The system was tested on multiple discussion scenarios consisting of group conversations with varying lengths and complexity. Each test case included multiple users exchanging messages related to decision-making, task assignment, and general discussion.

The evaluation parameters used in this project include:

- Accuracy of extracted summaries
- Relevance of identified decisions
- Correctness of action item detection
- System response time
- User interface usability

Table 6.1: Test Cases and Results

Test Case	Input Type	Expected Output	System Output	Result
TC-01	Short Discussion (5 msgs)	1 Decision, 1 Action Item	1 Decision, 1 Action Item	Pass
TC-02	Medium Discussion (15 msgs)	3 Decisions, 4 Action Items	3 Decisions, 3 Action Items	Partial
TC-03	Long Discussion (30+ msgs)	Summary + 5 Decisions	Summary + 4 Decisions	Partial
TC-04	Ambiguous Messages	No clear decisions	1 Incorrect Decision	Fail
TC-05	Mixed Conversation	2 Decisions, 2 Actions	2 Decisions, 2 Actions	Pass
TC-06	Simple Q&A Discussion	0 Decisions, 1 Action	0 Decisions, 1 Action	Pass

The results demonstrate that the system is capable of converting unstructured discussions into structured outputs effectively. It performs well for clearly structured conversations and shows expected limitations for highly ambiguous or informal messages.

6.2 Discussion

6.2.1 Performance of Basic System

The initial implementation of the system successfully processes user discussions and generates summaries. The summaries capture the key points of conversations and reduce the need for users to manually read long message threads. The system performs well for structured and moderately complex discussions where the intent of messages is clear.

6.2.2 Decision and Action Extraction

The system demonstrates the ability to identify decisions and action items from discussions. It detects sentences that indicate final decisions, task assignments, and responsibilities. This functionality improves clarity in collaborative environments and helps users track important outcomes.

6.2.3 System Limitations in Analysis

During testing, it was observed that the accuracy of extraction depends on the clarity of user messages. Ambiguous or informal conversations sometimes reduce the effectiveness of the NLP module. Additionally, complex discussions with mixed contexts may require more advanced models for better accuracy.

6.2.4 User Experience and Usability

The user interface provides a clear and structured view of results. Users can easily view summaries, track decisions, and identify action items. This improves usability and enhances the overall experience of the system.

6.3 Findings

The implementation of the system leads to several important findings:

31. Structured Output Improves Productivity: Converting discussions into summaries and action items significantly reduces manual effort.
32. AI Enhances Collaboration: The use of NLP techniques helps in extracting meaningful insights from conversations.
33. System Works Best with Clear Input: The performance of the system depends on the clarity and structure of input messages.
34. Prototype Demonstrates Practical Feasibility: Even a basic implementation shows strong potential for real-world applications.
35. Scalability is Achievable: The modular design allows future improvements and scalability of the system.

CHAPTER 7: Summary and Conclusion

The project was designed to develop an intelligent collaborative system capable of analyzing group discussions and extracting structured insights such as summaries, decisions, and action items. In modern digital environments, communication platforms generate large volumes of unstructured data, making it difficult for users to identify key outcomes.

The main objective of the project was to bridge this gap by integrating Natural Language Processing techniques into a collaborative platform. The system was designed using a modular architecture consisting of frontend, backend, and analysis components.

The implementation involved developing a web-based system where users can create teams, participate in discussions, and analyze conversations. The NLP module processes discussion data and generates meaningful outputs that improve clarity and productivity.

The results demonstrate that the system is capable of generating concise summaries of discussions, identifying decisions and responsibilities, and improving communication efficiency. The project also highlights the importance of applying AI techniques to real-world problems. By converting unstructured data into structured insights, the system enhances decision-making and collaboration.

In conclusion, the project successfully demonstrates the feasibility of integrating AI into collaborative systems. The developed prototype provides a strong foundation for future improvements and has potential applications in academic, professional, and organizational environments.

CHAPTER 8: Limitation of the Project and Future Work

8.1 Limitations of the Project

Although the project demonstrates promising results, several limitations must be considered:

36. Basic NLP Implementation: The current system uses basic NLP techniques, which may not handle highly complex or ambiguous conversations effectively.
37. Limited Dataset for Testing: The system was tested using sample datasets rather than large-scale real-world data.
38. No Real-Time Processing: The current implementation processes discussions after input rather than in real time.
39. Limited Context Understanding: The system may struggle with long conversations involving multiple topics.
40. Scalability Constraints: The current prototype is designed for small-scale usage and may require optimization for large user bases.

8.2 Future Work

The project can be further enhanced in several ways:

41. Integration of Advanced AI Models: Future versions can incorporate large language models for improved accuracy and understanding.
42. Real-Time Analysis: Implementing real-time processing of discussions can enhance usability.
43. Integration with Existing Platforms: The system can be integrated with tools like Slack or Microsoft Teams.
44. Improved Context Understanding: Advanced models can be used to better handle long and complex discussions.
45. Scalable Architecture: The system can be extended to support a larger number of users and discussions.
46. Enhanced Visualization: More advanced dashboards and analytics can be added for better insights.

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